Project Report

1. Problem Definition & Real-World Context

1.1 Research Question

How do median household income, population size, and median age at the ZIP-code level influence the rate of electric vehicle adoption (BEVs and PHEVs) across Washington State?

1.2 Real-World Context

Electric vehicles are a key component of Washington State's strategy to reduce carbon emissions and improve transportation sustainability. However, EV adoption is not uniform across communities. Understanding the socioeconomic and demographic factors that drive or limit EV adoption is important for designing equitable transportation policy.

This project investigates whether income level, population size, and median age at the ZIP-code level are meaningful predictors of EV adoption rates. Findings can help policymakers identify underserved communities and allocate infrastructure investment and incentive programs more effectively.

1.3 Modeling Task

This is a supervised regression problem. The target variable is 'ev_adoption_rate', defined as the number of registered EVs divided by the total population in each ZIP code. This normalization allows fair comparison across ZIP codes of different sizes.

2. Data Description & Preparation

2.1 Primary Dataset

The primary dataset is the Electric Vehicle Population Data (2025) from the Washington State Department of Licensing via Data.gov. It contains approximately 276,828 records of registered BEVs and PHEVs across Washington State ZIP codes. Key variables include ZIP code, vehicle type (BEV or PHEV), make, model, model year, and electric range.

2.2 Secondary Dataset

Supplementary demographic data were obtained from Data Commons using U.S. Census-derived indicators. This dataset provides ZIP code-level information on population size, median age, median household income, and median person income, covering 583 ZIP codes in Washington State.

2.3 Data Merging

The two datasets were merged using ZIP code as the common geographic identifier. A left join was applied to preserve all EV records from the primary dataset, even where demographic data was missing. ZIP codes were standardized to 5-character string format to ensure consistent matching.

2.4 Data Cleaning Steps

- Column names standardized to lowercase with underscores for consistency
- Duplicate records checked and removed and no duplicates found
- Irrelevant columns removed: VIN, legislative district, census tract, vehicle location, and DOL vehicle ID
- Records with missing median household income dropped (~9,000+ records)
- Data types corrected: population and income converted to integer; categorical columns encoded as category type
- VIN numbers partially masked (first and last characters replaced with 'X') to reduce privacy risk
- EV adoption rate recalculated as ev_count divided by population at the ZIP-code level
- Records with zero or null population removed to prevent division errors

2.5 Final Dataset Structure

The final cleaned dataset contains ZIP-code-level records with the following key variables: zip_code, county, city, median_income_household, population, median_age, ev_count, and ev_adoption_rate.

3. Modeling Approach & Design Choices

3.1 Feature Selection

Four features were selected based on the research question and data availability:

- median_income_household — primary socioeconomic predictor
- population — measure of ZIP-code size
- median_age — demographic predictor
- zip_code — geographic identifier (treated as categorical)

These variables directly match the research question and were consistently available across ZIP codes. Missing values in numeric features were imputed using the median strategy. ZIP code was encoded using one-hot encoding to capture geographic variation.

3.2 Baseline Model (Linear Regression)

Linear Regression was selected as the baseline model because it is simple, interpretable, and appropriate for continuous target variables. It helps quantify the linear relationship between each socioeconomic variable and EV adoption rate. Coefficients from Linear Regression directly indicate the direction and relative magnitude of each feature's influence.

3.3 Improved Model (Random Forest Regressor)

A Random Forest Regressor was trained as the improved model. Random Forest is an ensemble method that builds multiple decision trees and averages their predictions. It was selected because EV adoption patterns may involve non-linear relationships and interaction effects between variables that Linear Regression cannot capture. The model used 200 estimators with no maximum depth constraint.

3.4 Preprocessing Pipeline

A scikit-learn Pipeline was used to ensure consistent preprocessing for both models. Numeric features were imputed using median values. Categorical features (zip_code) were imputed with the most frequent value and encoded using OneHotEncoder with handle_unknown set to ignore. This pipeline prevents data leakage by applying transformations only after train-test splitting.

3.5 Train-Test Split

The dataset was split into 80% training and 20% test sets using a fixed random seed (random_state=42) for reproducibility. Importantly, a single split was applied and shared across both models to ensure fair comparison.

4. Results & Evaluation

4.1 Model Performance Comparison

The following table summarizes the evaluation metrics for both models on the held-out test set:

Model	MAE	RMSE	R^2
Linear Regression	0.02	0.03	0.236
Random Forest	0.02	0.05	−1.883

4.2 Baseline Model (Linear Regression)

The Linear Regression model achieved an MAE of 0.02, meaning predictions differ from actual adoption rates by approximately 2 percentage points on average. The RMSE of 0.03 indicates some larger deviations exist but remain relatively small in absolute terms. The R^2 value of 0.236 means that approximately 23.6% of the variation in EV adoption rates is explained by median income, population, and median age.

4.3 Improved Model (Random Forest)

The Random Forest model produced the same MAE of 0.02 but a higher RMSE of 0.05, suggesting it makes larger prediction errors on some ZIP codes. More critically, its R^2 value of −1.883 indicates that it performs significantly worse than simply predicting the mean adoption rate for all ZIP codes. This negative R^2 result was initially caused by a data leakage bug where train-test split was applied twice, training Random Forest on mismatched data. After fixing this, the negative R^2 persisted, indicating that Random Forest overfits the training data and fails to generalize.

4.4 Cross-Validation Results

Five-fold cross-validation was applied to both models to provide a more reliable performance estimate. Results showed high variance across folds for both models, with particularly extreme negative R^2 values in Fold 5 for both. This instability confirms that the current feature set does not generalize well across different subsets of ZIP codes, likely due to influential outlier ZIP codes such as Grays Harbor and San Juan appearing in some folds but not others.

4.5 Feature Importance & Coefficients

Linear Regression coefficients show that median_age has the largest positive coefficient, meaning older ZIP codes are associated with slightly higher EV adoption rates when controlling for other variables. median_income_household and population had near-zero coefficients, suggesting weak linear associations in isolation.

Random Forest feature importance scores show a different pattern: zip_code (37%) and population (32%) are the most important features, followed by median_income_household (23%), and median_age (8%). This divergence between models confirms that EV adoption involves complex non-linear relationships that vary by geographic location.

4.6 Geographic Summary

County-level analysis reveals a clear geographic divide. King County leads with the highest mean EV adoption rate (7.68%), followed by San Juan (7.21%) and Grays Harbor (7.10%). Eastern counties such as Franklin (0.65%), Pend Oreille (0.66%), and Yakima (0.71%) show the lowest adoption rates.

The county-level income versus adoption rate scatter plot shows a general positive trend, but notable outliers like Grays Harbor which has a relatively low income but a high adoption rate indicate that income is not the sole driver of EV adoption.

5. Errors, Limitations & Ethical Considerations

5.1 Model Errors & Technical Limitations

- Limited explanatory variables: only income, population, median age, and ZIP code are included. Key missing variables include charging infrastructure availability, housing type (single-family vs. multi-unit), urban density, local EV incentives, and total number of registered vehicles per ZIP code.
- Cross-sectional data: the analysis uses data from a single time period. EV adoption is a dynamic process and trends may have shifted since data collection.
- ZIP-code aggregation: aggregating data at the ZIP-code level may hide socioeconomic inequality within the same area, known as the ecological fallacy.
- Small ZIP-code instability: ZIP codes with very small populations may show extreme adoption rates that are not representative of broader trends.
- Data leakage bug (corrected): the original code applied train-test split twice, causing the Random Forest to train on incorrect data. This was corrected in the final version.

5.2 Potential Bias

- Socioeconomic bias: because EVs are relatively expensive, income is strongly correlated with EV adoption. Using income as a predictor may reinforce existing inequality patterns if findings are used to allocate resources.
- Urban overrepresentation: high-income urban areas such as King and Snohomish counties contribute a large share of EV registrations, which may skew model learning toward urban patterns.
- Missing data removal: approximately 9,000+ records with missing median household income were removed, potentially underrepresenting rural or lower-income ZIP codes in the analysis.

5.3 Ethical Considerations

This study aims to understand disparities in EV adoption, not to label or rank communities. Results should be interpreted carefully and responsibly:

- Lower EV adoption rates do not reflect lack of interest or awareness. Structural barriers including affordability, charging infrastructure, housing type, and access to financing may limit adoption in rural and lower-income communities.
- Statistical associations identified in this study do not imply causation. The relationship between income and EV adoption may be mediated by many unmeasured factors.
- Policymakers should use these findings to improve transportation equity — for example, by expanding public charging infrastructure in underserved areas and expanding EV incentive programs — rather than to penalize communities with low adoption rates.

- All data is analyzed at the ZIP-code level. No personally identifiable information is included in the final analysis dataset.

6. Final Conclusion & Best Model Selection

Based on all evaluation results, Linear Regression is selected as the final and better-performing model for this dataset. Although neither model achieved strong predictive performance in absolute terms, Linear Regression is more stable and interpretable. It achieves a positive R^2 of 0.236 on the test set, while Random Forest produces a severely negative R^2 of −1.883, indicating that it performs worse than simply predicting the average adoption rate for all ZIP codes.

Cross-validation confirms that both models are highly unstable across folds, particularly in Fold 5 where both models collapse. This instability is driven by outlier ZIP codes whose adoption rates are not well explained by the available features. Random Forest shows more severe instability, particularly due to its tendency to overfit when geographic identifiers like ZIP code dominate feature importance.

The low R^2 across both models clearly indicates that median income, population size, and median age alone are insufficient to explain EV adoption patterns across Washington State. The model explains roughly one-quarter of the variation, with the remaining three-quarters attributable to factors not captured in this analysis.

Future work should incorporate additional variables such as charging station density, housing type, local EV purchase incentives, commuting patterns, and vehicle registration per capita. These additions are expected to significantly improve model performance and provide more actionable insights for transportation equity planning across Washington State.